

Common Name: ISOPROPYL ALCOHOL (IPA) 50%, 70%, 91% AND 99% IPA
Manufacturer: CUMBERLAND-SWAN
MSDS Revision Date: 7/1/2006

SECTION 1: CHEMICAL PRODUCT AND COMPANY IDENTIFICATION

MANUFACTURER:
CUMBERLAND SWAN
ONE SWAN DRIVE
SMYRNA, TN 37167

DATE: JULY 2006

PRODUCT: ISOPROPYL ALCOHOL (IPA) 50%, 70%, 91% AND 99% IPA

TELEPHONE: (615) 459-8900
24HR EMERGENCY: (615) 459-8900 EXT. 5270

SECTION 2: COMPOSITION / INFORMATION ON INGREDIENTS

NAME:

ISOPROPANOL, IPA, 2-PROPANOL, DIMETHYL CARBINOL CAS#: 67-63-0

DEIONIZED WATER

SECTION 3: HAZARDS IDENTIFICATION

COLORLESS, VOLATILE LIQUID WITH THE ODOR OF RUBBING ALCOHOL. ISOPROPYL ALCOHOL IS A DANGEROUS FIRE RISK. PROLONGED EXPOSURE TO ELEVATED CONCENTRATIONS OF VAPORS MAY RESULT IN IRRITATION OF THE EYES, NOSE, AND THROAT AND CENTRAL NERVOUS SYSTEM (CNS) DEPRESSION. PROLONGED DERMAL EXPOSURE CAN RESULT IN DRY, CRACKING SKIN.

POTENTIAL ROUTES OF EXPOSURE:
INGESTION, INHALATION, DERMAL CONTACT, EYE CONTACT

TARGET ORGANS: EYES, SKIN, RESPIRATORY SYSTEM

SYMPTOMS OF OVEREXPOSURE:

INHALATION: MILD IRRITATION OF EYES, NOSE AND THROAT.

INGESTION: DROWSINESS, HEADACHE

DERMAL CONTACT: DRY, CRACKING SKIN

ACUTE EFFECTS:

IRRITATION OF SKIN AND/OR UPPER RESPIRATORY TRACT AS NOTED ABOVE.

ACUTE CNS DEPRESSION MAY BE MANIFESTED AS GIDDINESS, HEADACHE, DIZZINESS AND/OR NAUSEA.

CHRONIC EFFECTS:

CHRONIC EXPOSURE CAN RESULT IN SKIN IRRITATION AND CONTACT DERMATITIS
PRE-EXISTING DISORDERS OF THE SKIN, EYES, AND RESPIRATORY TRACT MAY BE
EXACERBATED BY EXPOSURE TO ISOPROPYL ALCOHOL.

HMIS:

H=1

F=3

R=0

SEE SECTION 8 FOR PPE INFORMATION

SECTION 4: FIRST AID MEASURES

EYE: FLUSH EYES WITH COPIOUS AMOUNT OF WATER FOR AT LEAST 15 MINUTES

SKIN: FLUSH WITH WATER. IF IRRITATION PERSISTS, SEEK MEDICAL ATTENTION.

INGESTION:

DO NOT INDUCE VOMITING IF VICTIM IS UNCONSCIOUS OR DROWSY. SEEK MEDICAL
ATTENTION OR CONTACT THE POISON CONTROL CENTER.

INHALATION:

REMOVE VICTIM TO FRESH AIR AND PROVIDED OXYGEN IF BREATHING IS DIFFICULT.
SEEK MEDICAL ATTENTION IF BREATHING CONTINUES TO BE DIFFICULT.

SECTION 5: FIRE FIGHTING MEASURES

EXTINGUISHING MEDIA: USE WATER FOG, ALCOHOL FOAM, DRY CHEMICAL OR CO2

UNUSUAL FIRE OR EXPLOSION HAZARDS:

CONTAINERS EXPOSED TO INTENSE HEAT FROM FIRES SHOULD BE COOLED WITH LARGE
AMOUNTS OF WATER TO PREVENT BUILDUP OF INTERNAL PRESSURE DUE TO VAPOR
GENERATION WHICH COULD RESULT IN CONTAINER RUPTURE.

RECOMMENDATIONS:

CLEAR AREA OF UNPROTECTED PERSONNEL. WEAR COMPLETE TURNOUT GEAR. COOL
CONTAINERS EXPOSED TO FIRE WITH WATER.

SECTION 6: ACCIDENTAL RELEASE MEASURES

LARGE SPILLS:

ELIMINATE ALL IGNITION SOURCES. EQUIPMENT MUST BE GROUNDED TO PREVENT
SPARKING. EVACUATE THE AREA OF UNPROTECTED PERSONNEL. CONTAIN SOURCE OF

SPILL. DIKE OR OTHERWISE CONFINE SPILLED PRODUCT. UNCONTROLLED RELEASES TO AIR, LAND, OR WATER MAY BE REPORTABLE TO THE NATIONAL RESPONSE CENTER (1-800-424-8802).

SMALL SPILLS:

TAKE UP WITH ABSORBENT MATERIAL AND PLACE IN NON-LEAKING CONTAINER; SEAL TIGHTLY. DISPOSE OF ABSORBENT (SEE SECTION 13)

SECTION 7: HANDLING AND STORAGE

STORAGE REQUIREMENTS:

STORE IN TIGHTLY CLOSED CONTAINERS IN A COOL, DRY AREA AWAY FROM HEAT AND OTHER POSSIBLE IGNITION SOURCES.

HANDLING PRECAUTIONS:

USE NON-SPARKING TOOLS TO OPEN CONTAINERS. MAINTAIN APPROPRIATE CLASS OF FIRE EXTINGUISHERS NEARBY IN CASE OF FIRE.

SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

OSHA PEL=400PPM

OSHA STEL=500PPM

IDLH=12,000PPM

RECOMMENDED ENGINEERING CONTROLS:

USE EXPLOSION-PROOF VENTILATION EQUIPMENT AS NECESSARY TO MAINTAIN AIRBORNE CONCENTRATIONS BELOW THE PEL.

GROUND ALL CONTAINERS TO PREVENT STATIC SPARKS DURING FLUID TRANSFERS.

RECOMMENDED ADMIN CONTROLS:

TRAIN EMPLOYEES ON THE HAZARDS OF ISOPROPYL ALCOHOL

PPE:

GOGGLES, GLOVES, NIOSH APPROVED RESPIRATORY PROTECTION REQUIRED WHEN ABOVE PEL/TWA

RECOMMENDED HYGIENE PRACTICES:

CLEAN PPE AND WORK CLOTHING CONTAMINATED PRIOR TO REUSE. AFTER WORKING WITH THIS PRODUCT, BE SURE TO WASH BEFORE EATING, SMOKING, DRINKING, OR APPLYING COSMETICS.

SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

APPEARANCE: COLORLESS LIQUID

UEL: 12%

LEL: 2%

ODOR: MILD RUBBING ALCOHOL

ODOR THRESHOLD: 43PPM

WATER SOLUBILITY: MISCIBLE

50% IPA 70%IPA 91%IPA 99%IPA

VAPOR PRESSURE (@ 68 DEG. F) APPROX. 29MM 23MM 33MM 33MM

SPECIFIC GRAVITY .929 .878 .790 .790

BOILING POINT 176 176 180 181
DEG. F DEG. F DEG. F DEG. F

FLASH POINT (TAG OPEN CUP) 74.5 70.5 54 53
DEG. F DEG. F DEG. F DEG. F

FREEZING POINT 32-50 32-50 32-50 127
DEG. C DEG. C DEG. C DEG. F

MOLECULAR WEIGHT 47.5 47.5 47.5 60.1

AUTO IGNITION TEMPERATURE NO DATA NO DATA NO DATA 750 DEG. F

SECTION 10: STABILITY AND REACTIVITY

STABILITY: STABLE

POLYMERIZATION: WILL NOT OCCUR

INCOMPATIBLE CHEM:
STRONG OXIDIZERS, ACETALDEHYDE, CHLORINE, ETHYLENE OXIDE, ACIDS, ISOCYANATES

CONDITIONS TO AVOID:
HEAT, SPARKS, AND OPEN FLAME.
DO NOT STORE IN ALUMINUM > 120 DEG. F

HAZARDOUS PRODUCTS:
CO AND UNIDENTIFIED ORGANIC COMPOUNDS MAY BE FORMED OF DECOMPOSITION

SECTION 11: TOXICOLOGICAL INFORMATION

LD50:
5,840 MG/KG (ACUTE ORAL - RAT)
13,000 MG/KG (ACUTE DERMAL - RABBIT)

LD50: 16,000 PPM/8HR (INHALATION - RAT)

LDL0: 5,000 MG/KG (ORAL - RABBIT)

MUTAGENICITY: NOT INDICATED

REPRODUCTIVE EFFECTS: NOT INDICATED

CARCINOGENICITY: NOT IDENTIFIED AS A CARCINOGEN BY OSHO, IARC, OR NTP

SECTION 12: ECOLOGICAL INFORMATION

ECOTOXICITY: N/A

ENVIRONMENTAL FATE: N/A

SOIL ABSORPTION/MOBILITY: HIGHLY MOBILE

ENVIRONMENTAL DEGRADATION:
SHOULD BE REMOVED READILY FROM SOILS AND WATER BY VOLATILIZATION AND
BIODEGRADATION.

SECTION 13: DISPOSAL CONSIDERATIONS

DISPOSAL:
CONTACT YOUR SUPPLIER OR A LICENSED CONTRACTOR FOR DETAILED RECOMMENDATIONS.

DISPOSAL REGULATORY REQUIREMENTS:
FOLLOW APPLICABLE FEDERAL, STATE, AND LOCAL REGULATIONS. CONSIDER FUELS
BLENDING AS AN ALTERNATIVE TO INCINERATION.

SECTION 14: TRANSPORT INFORMATION

DOT SHIPPING NAME: ISOPROPANOL
DOT PACKING GROUP: II
DOT HAZARD CLASS: 3
DOT LABEL: FLAMMABLE LIQUID
UN ID#: UN 1219

OTHER REGULATED MATERIALS - DOMESTIC (ORM-D) - NOT DOT REGULATED (49 CFR
173.144); QUANTITIES NOT GREATER THAN 1 LITER OR 0.3 GALLONS AND PGII MAY BE
SHIPPED AS ORM-D (49 CFR 173.150) AS A CONSUMER COMMODITY (49 CFR 171.8).

SECTION 15: REGULATORY INFORMATION

RCRA HAZARDOUS WASTE NUMBER/CLASSIFICATION:
D001 CERCLA SUBSTANCE: N/A

HAZARDOUS AIR POLLUTANT (CAA): NO

SARA 311/312 CODES: N/A

SARA TOXIC CHEMICAL: YES, (STRONG MANUFACTURING ONLY)

CERCLA REPORTABLE QUANTITY: 10,000 LBS (DEFAULT)

TSCA: THE INGREDIENTS OF THIS PRODUCT ARE ON THE TSCA INVENTORY.

SECTION 16: OTHER INFORMATION

PREPARED BY: CUMBERLAND SWAN

SOURCES OF INFORMATION:

29 CFR1910.1000; NIOSH POCKET GUIDE TO CHEMICAL HAZARDS (1993); OCCUPATIONAL HEALTH GUIDELINES FOR CHEMICAL HAZARDS; NFPA GUIDE TO HAZARDOUS MATERIALS - 10TH EDITION.

DISCLAIMER:

WHILE REASONABLE CARE HAS BEEN TAKEN TO ENSURE THE ACCURACY AND COMPLETENESS OF THE INFORMATION REGARDING THE MATERIAL DESCRIBED HEREIN, IT IS THE PURCHASER'S RESPONSIBILITY TO ENSURE THE SUITABILITY OF SUCH INFORMATION AS IT APPLIES TO THE PURCHASER'S INTENDED USE OF THE MATERIAL.

ISOPROPYL ALCOHOL MSDS